# BACKPLANE / CONNECTION BLOCKS

PCB-oriented top view | Continuous wires | Test points omitted | 21 Oct 2026

## U2D2 / DYNAMIXEL

M1 / U2D2

USB to host computer

Mount: no PCB signal pads

J7 / SERVO BUS

DYNAMIXEL_DATA

+12V

GND

## ARDUINO MEGA / TEENSY ADAPTER

UNUSED CONTACTS / NC

A1 / MEGA-PATTERN SOCKET

Alternative: Teensy on adapter

A5 FEMALE_1_/_LIGHT_SENSOR
A6 FEMALE_2_/_LIGHT_SENSOR
A7 FEMALE_3_/_LIGHT_SENSOR
A9 MALE_1_/_LIGHT_SENSOR_/_B
A11 MALE_1_/_LIGHT_SENSOR_/_D
A8 MALE_1_/_LIGHT_SENSOR_/_A
A10 MALE_1_/_LIGHT_SENSOR_/_C
A13 MALE_2_/_LIGHT_SENSOR_/_B
A15 MALE_2_/_LIGHT_SENSOR_/_D
A12 MALE_2_/_LIGHT_SENSOR_/_A
A14 MALE_2_/_LIGHT_SENSOR_/_C
A0 FEMALE_1_/_ANALYSER_OUT
A1 FEMALE_2_/_ANALYSER_OUT
A2 FEMALE_3_/_ANALYSER_OUT
A3 MALE_1_/_ANALYSER_OUT
A4 MALE_2_/_ANALYSER_OUT
D26 FEMALE_1_/_MIC_DIRECT
D27 FEMALE_2_/_MIC_DIRECT
D28 FEMALE_3_/_MIC_DIRECT
D29 MALE_1_/_MIC_DIRECT
D30 MALE_2_/_MIC_DIRECT

D6 FEMALE_1_/_TONE
D46 FEMALE_2_/_TONE
D10 FEMALE_3_/_TONE
D11 MALE_1_/_TONE
D5 MALE_2_/_TONE
D4 ANALYSER_/_STROBE
D3 ANALYSER_/_RESET
D14 FEMALE_1_/_NEOPIXEL_DATA_/_DRIVEN
D7 FEMALE_2_/_NEOPIXEL_DATA_/_DRIVEN
D8 FEMALE_3_/_NEOPIXEL_DATA_/_DRIVEN
D9 MALE_1_/_NEOPIXEL_DATA_/_DRIVEN
D15 MALE_2_/_NEOPIXEL_DATA_/_DRIVEN
D16 MALE_1_/_BAR_NEOPIXEL_DATA_/_DRIVEN
D17 MALE_2_/_BAR_NEOPIXEL_DATA_/_DRIVEN
D2 AMP_SHUTDOWN
D24 MALE_1_/_AUX_DRIVEN
D25 MALE_2_/_AUX_DRIVEN

GND + MEGA_5V contacts: GND1 5V1 5V2 5V3 5V4 GND2 GND3 GND4 GND5 GND6

## POWER ENTRY / SUPPLIES

J2 / 5 V INPUT

+5V
GND

J6 / SCREW BRIDGE

GND
GND
+5V
GND
+5V
+5V

C1 470uF 25V
C2 100nF

JP1 OR

AGND / GND bond

RLP5 2K2
RLP12 4K7
RLPUSB 2K2

DP5
DP12
DPUSB

## LIGHT SENSOR LOADS

RP1 10K provisional
RP2 10K provisional
RP3 10K provisional
RP4 10K provisional
RP5 10K provisional
RP6 10K provisional
RP7 10K provisional
RP8 10K provisional
RP9 10K provisional
RP10 10K provisional
RP11 10K provisional

## LED DATA / AUX OUTPUTS

RS1 10K

AMP SHUTDOWN PULLUP

RN1 330R
RN2 330R
RN3 330R
RN4 330R
RN5 330R
RN6 330R
RN7 330R
RA1 330R
RA2 330R

RND1 100K
RND2 100K
RND3 100K
RND4 100K
RND5 100K
RND6 100K
RND7 100K

## J1 / FEMALE 2

0 GND
1 FEMALE 2 / SPARE1
2 FEMALE 2 / SPARE2
3 FEMALE 2 / SPARE3
4 FEMALE 2 / AUDIO RETURN
5 FEMALE 2 / LIGHT SENSOR
6 FEMALE 2 / MIC
7 +12V
8 GND
9 FEMALE 2 / SPARE4
10 FEMALE 2 / SPARE5
11 FEMALE 2 / SPARE6
12 FEMALE 2 / LINE OUT
13 FEMALE 2 / NEOPIXEL DATA
14 DYNAMIXEL DATA
15 +5V

## J5 / FEMALE 1

0 GND
1 FEMALE 1 / SPARE1
2 FEMALE 1 / SPARE2
3 FEMALE 1 / SPARE3
4 FEMALE 1 / AUDIO RETURN
5 FEMALE 1 / LIGHT SENSOR
6 FEMALE 1 / MIC
7 +12V
8 GND
9 FEMALE 1 / SPARE4
10 FEMALE 1 / SPARE5
11 FEMALE 1 / SPARE6
12 FEMALE 1 / LINE OUT
13 FEMALE 1 / NEOPIXEL DATA
14 DYNAMIXEL DATA
15 +5V

## A-J3 / FEMALE 3 + MALE 1

0 GND
1 GND
2 +12V
3 FEMALE 3 / MIC
4 FEMALE 3 / LIGHT SENSOR
5 FEMALE 3 / AUDIO RETURN
6 MALE 1 / NEOPIXEL DATA
7 MALE 1 / LIGHT SENSOR / B
8 MALE 1 / LIGHT SENSOR / D
9 +5V
10 DYNAMIXEL DATA
11 FEMALE 3 / NEOPIXEL DATA
12 FEMALE 3 / LINE OUT
13 MALE 1 / MIC
14 MALE 1 / LIGHT SENSOR / A
15 MALE 1 / LIGHT SENSOR / C

## B-J4 / MALES + BAR

0 GND
1 MALE 1 / AUX
2 MALE 1 / AUDIO RETURN
3 MALE 2 / NEOPIXEL DATA
4 MALE 2 / LIGHT SENSOR / B
5 MALE 2 / LIGHT SENSOR / D
6 MALE 2 / LINE OUT
7 CENTRE / SPARE1
8 MALE 2 / BAR NEOPIXEL DATA
9 MALE 1 / LINE OUT
10 MALE 2 / MIC
11 MALE 2 / LIGHT SENSOR / A
12 MALE 2 / LIGHT SENSOR / C
13 MALE 2 / AUX
14 MALE 2 / AUDIO RETURN
15 MALE 1 / BAR NEOPIXEL DATA

## VOICES / FIVE CARD SOCKETS

FEMALE 1 — JV1
FEMALE 2 — JV2
FEMALE 3 — JV3
MALE 1 — JV4
MALE 2 — JV5

+5V
TONE
AUDIO_OUT
GND
AGND

EXTRA1 / F2 SPARES

FEMALE_2_/_SPARE1
FEMALE_2_/_SPARE2
FEMALE_2_/_SPARE3
FEMALE_2_/_SPARE4
FEMALE_2_/_SPARE5
FEMALE_2_/_SPARE6

RLT1 1K5
RLT2 1K5
RLT3 1K5
RLT4 1K5
RLT5 1K5
R203 100R
DT1
DT2
DT3
DT4
DT5

R303 100R
R403 100R
R503 100R
R103 100R
RL1 100K
RL2 100K
RL3 100K
RL4 100K
RL5 100K

JP4 0R
JP5 0R
JP6 0R
JP2 0R
JP3 0R
HV1
HV2
HV3
HV4
HV5

EXTRA2 / F1 SPARES

FEMALE_1_/_SPARE1
FEMALE_1_/_SPARE2
FEMALE_1_/_SPARE3
FEMALE_1_/_SPARE4
FEMALE_1_/_SPARE5
FEMALE_1_/_SPARE6

## AUDIO ANALYSERS / FIVE CHANNELS

JA1

DFR0126 ANALYSER CARRIER

5 microphones -> 5 analogue outputs

Shared RESET / STROBE

MEGA_5V supply / AGND returns

1 FEMALE_1_/_MIC
2 AGND
11 FEMALE_1_/_ANALYSER_OUT
3 FEMALE_2_/_MIC
4 AGND
12 FEMALE_2_/_ANALYSER_OUT
5 FEMALE_3_/_MIC
6 AGND
13 FEMALE_3_/_ANALYSER_OUT
7 MALE_1_/_MIC
8 AGND
14 MALE_1_/_ANALYSER_OUT
9 MALE_2_/_MIC
10 AGND
15 MALE_2_/_ANALYSER_OUT
16 AGND
17 ANALYSER_/_STROBE
18 ANALYSER_/_RESET
19 MEGA_5V
20 GND
21 GND

HA1
HA2

## DIRECT MICROPHONE INPUTS

FEMALE 1 — RM1 4K7 — RMD1 1M
FEMALE 2 — RM2 4K7 — RMD2 1M
FEMALE 3 — RM3 4K7 — RMD3 1M
MALE 1 — RM4 4K7 — RMD4 1M
MALE 2 — RM5 4K7 — RMD5 1M

CENTRE SPARE

CENTRE_/_SPARE1

FEMALE 2 HARNESS
J1 / DSUB-15

FEMALE 1 HARNESS
J5 / DSUB-15

A-J3 / DSUB-15
FEMALE 3 / MALE 1 HARNESS

B-J4 / DSUB-15
MALE 1 / MALE 2 / BAR HARNESS

GND
AGND
+5V
+12V
MEGA_5V

AGND = ANALOG RETURN | MEGA_5V = CONTROLLER 5 V | Dots join wires; crossings without dots stay separate.